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Hydrodynamic singularities

When a drop or a bubble pinches off, the flow forgets its past. A pinch of polymer brings the memory back — but only for the drop.

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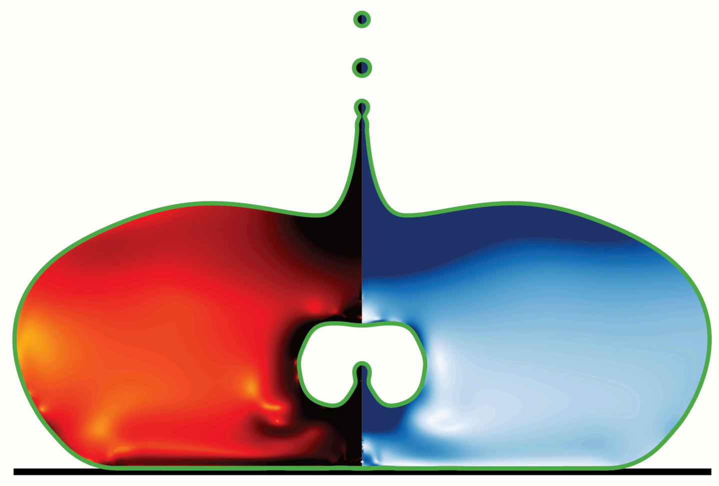
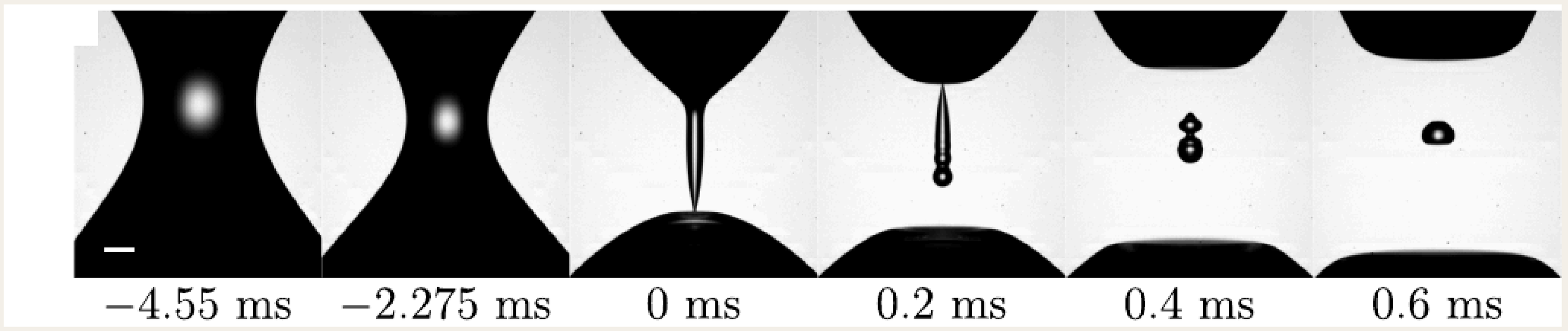


A **singularity** is when a smooth physical process blows up — something finite becomes infinite in a finite time or at a finite place. As a drop or a bubble pinches off, the neck radius h races to zero and the shape turns self-similar, $h \sim (t_0 - t)^{\alpha}$: a universal form set only by the local balance of inertia, surface tension and viscosity. The flow **forgets how it began**. Singularities erase a fluid's memory.

A **drop** pinches off

$h \sim (t_0 - t)^{2/3}$

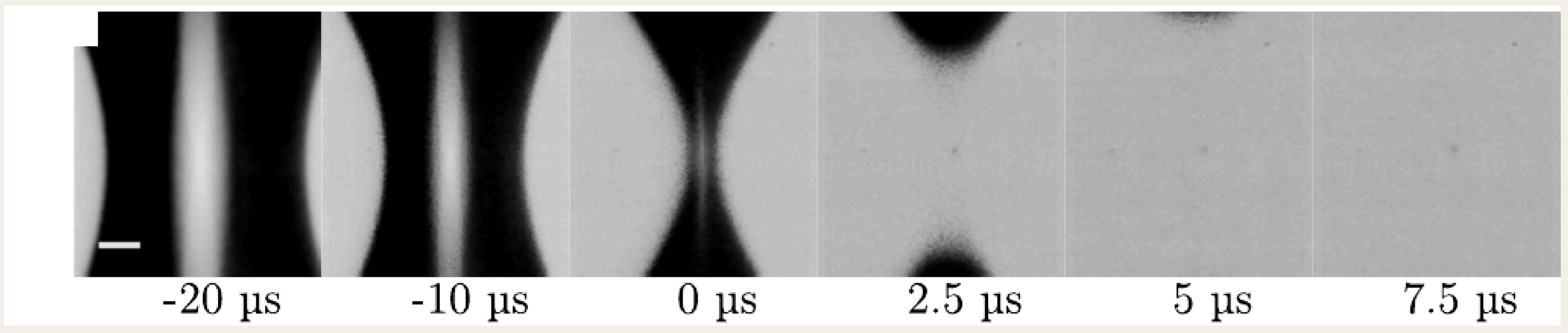
Plain water. Surface tension pulls a slender axial thread to a point, then it snaps into satellite drops — milliseconds.



A **bubble** pinches off

$h \sim (t_0 - t)^{1/2}$

Plain water. The surrounding liquid's inertia collapses the cavity radially — microseconds, a thousand times faster.



Two free surfaces, the same finite-time blow-up: read left to right, $t = 0$ is the singularity. They reach it by different routes (a drop along a thread, a bubble by collapsing a cavity), so they carry different exponents. **What if we gave the fluid a memory?**

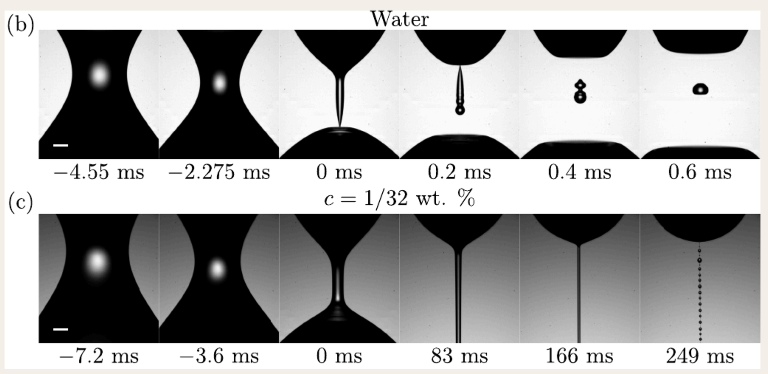
Elasticity is memory

Dissolve a few polymer chains and the liquid gains a memory — each chain stores its stretching history over a relaxation time λ. Push to the perfectly elastic limit ($\lambda \to \infty$, nothing is forgotten) and ask whether memory can outlast the singularity. It does for a drop, yet a dilute bubble pinches off as if the polymer were not there. Why the difference?

The drop remembers

A water drop reaches a genuine finite-time singularity. The pinch is capillary-driven, so the neck thins into a slender axial thread and the polymers stretch *along* it as it closes.

The elastic stress then diverges as $\sigma_{zz} \sim G\,(h_0/h)^4$ — faster than the capillary drive $\gamma\kappa \sim \gamma/h$ it fights. It must win: the collapse is arrested into a long-lived **beads-on-a-string** thread, and in the perfectly elastic limit the singularity is removed altogether.



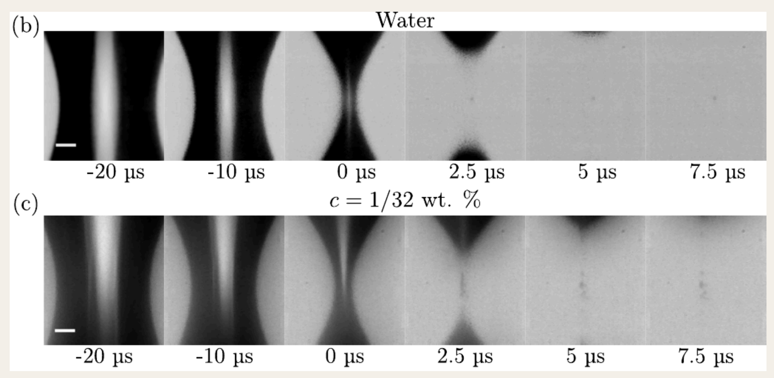
Drop pinch-off. Top: plain water reaches the singularity and snaps into satellites. Bottom: the same dilute polymer draws the neck into a persistent beads-on-a-string thread.

→ Dilute polymer → a thread. Memory wins.

The bubble forgets anyway

A bubble neck pinches just as sharply, $h \sim (t_0 - t)^{1/2}$, but the collapse is driven by the liquid's inertia and the cavity closes *radially*, so the polymers stretch sideways rather than along a thread.

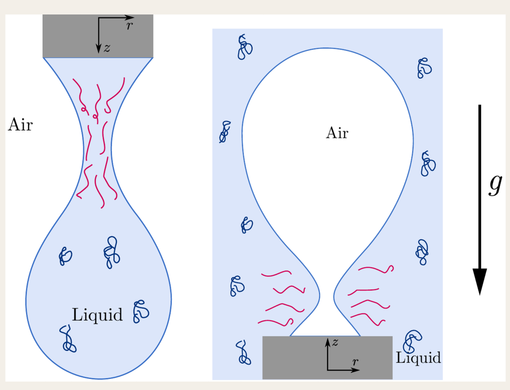
Now the elastic stress diverges only as $\sigma_{rr} \sim G\,(h_0/h)^2$ — the *same* rate as the inertial drive $\sigma_I \sim \rho\dot{h}^2$. A fair race it cannot win: it stays subdominant, so the bubble reaches its singularity **with or without memory**. A thread needs high concentration, and even then its fate hangs on the needle size.



Bubble pinch-off. Top: water. Bottom: the same dilute polymer. In the dilute limit the two near-singular sequences are almost indistinguishable.

→ Dilute polymer → no thread. The singularity survives.

Why elasticity picks sides



Polymers (blue coils) stretch where the neck thins (red): **along** the drop's axial thread, but only **sideways** across the bubble's neck.

	Drop	Bubble
WHAT DRIVES THE PINCH	surface tension, $\gamma\kappa \sim \gamma/h$	the liquid's inertia, $\sigma_I \sim \rho\dot{h}^2$
NECK GEOMETRY	a slender **axial** thread	a **radial** cavity collapse
ELASTIC STRESS BUILDS AS	$\sigma_{zz} \sim G(h_0/h)^4$	$\sigma_{rr} \sim G(h_0/h)^2$
...VS THE DRIVE → FATE	**faster** (4 vs 1): a thread; singularity removed	**same** rate (2 vs 2): it pinches; singularity survives

It is a race between the elastic stress and whatever drives the pinch: the drop's stress outgrows its capillary drive, the bubble's only ties its inertial drive. Same additive, opposite fate — and reading which singularities a touch of elasticity can rewrite is how we learn to control breakup: inkjet printing, sprays and aerosols.

References

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Acknowledgements

Simulations with **Basilisk C** (Popinet, *J. Comput. Phys.* **228**, 5838, 2009) on Durham's **Hamilton** HPC. A CoMPhy Lab (Durham) × Physics of Fluids (Twente) collaboration.

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